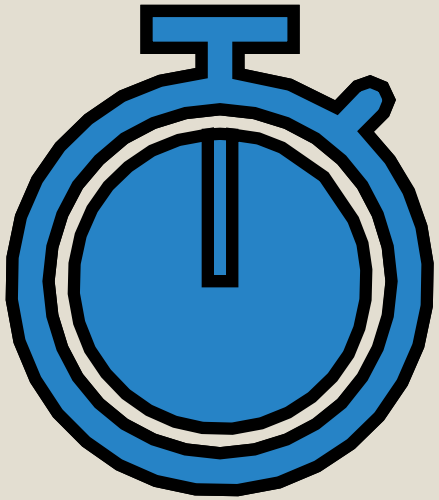
# AGILE SOFTWARE DEVELOPMENT

2022H1 OSE (Kasper, Shokoofeh)

# Learning Objectives

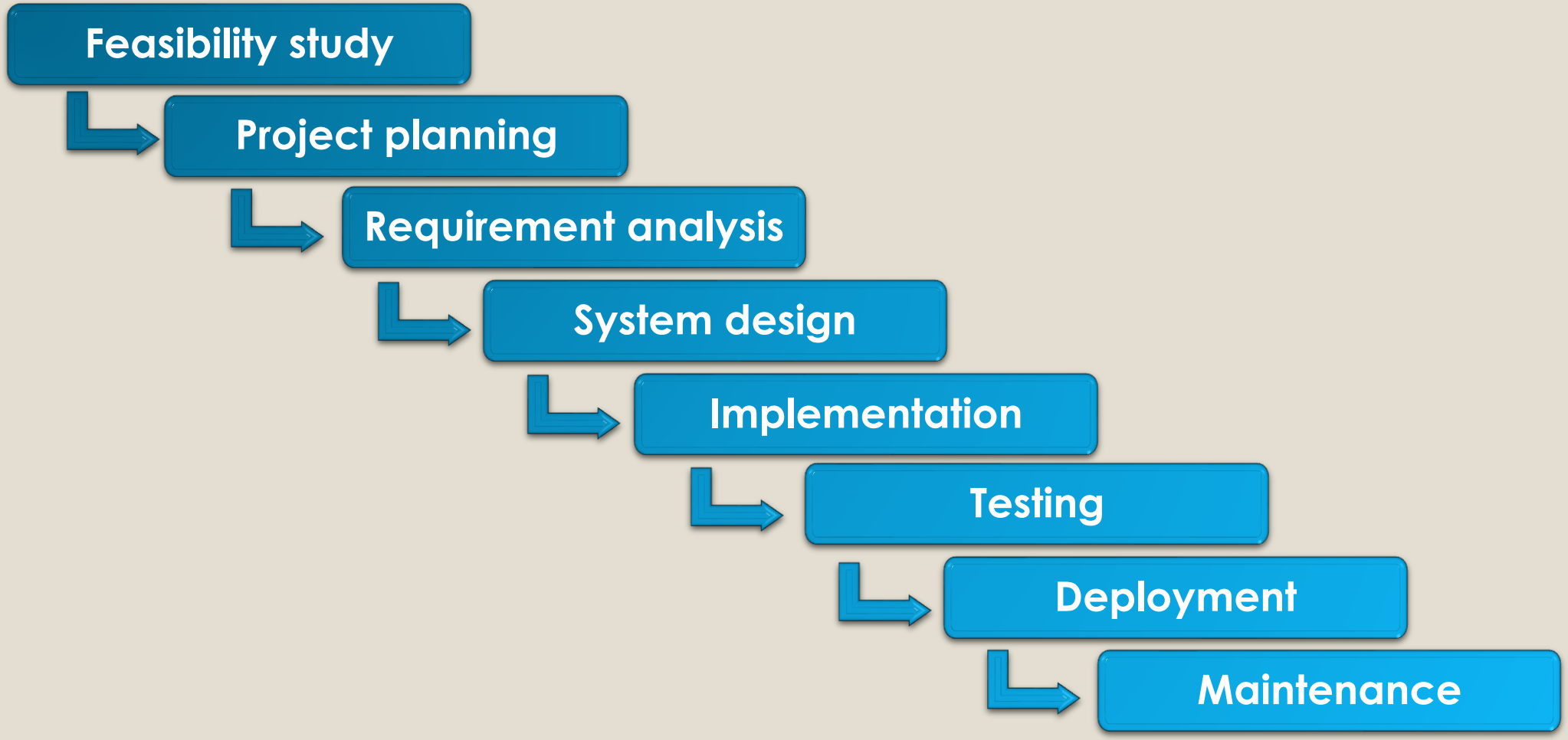
- Getting familiar with **agile** software development methodology;
- Getting familiar with **Scrum** framework

# Let’s learn together (5 min)



- Form groups of 5-7
- Choose a name for your group
- Write it down on a paper
- Each student can have a scrap paper
-

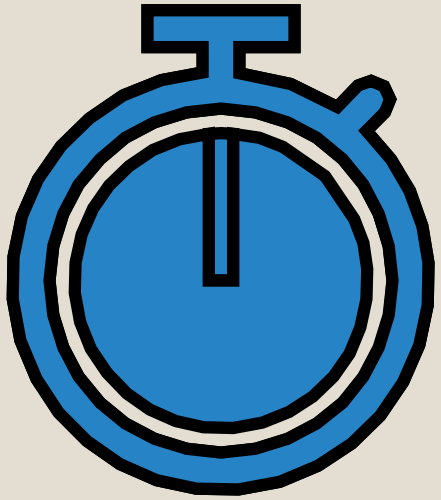
# Recap: SDLC


Feasibility study
Project planning
Requirement analysis
System design
Implementation
Testing
Deployment
Maintenance

# Restaurant : real-world example of SDLC

- Order a food
- Customize the order
- Requirement analysis
- System design
- Implementation
- Testing
- Order is ready
- Maintenance 🤮

# Real-world example of SDLC

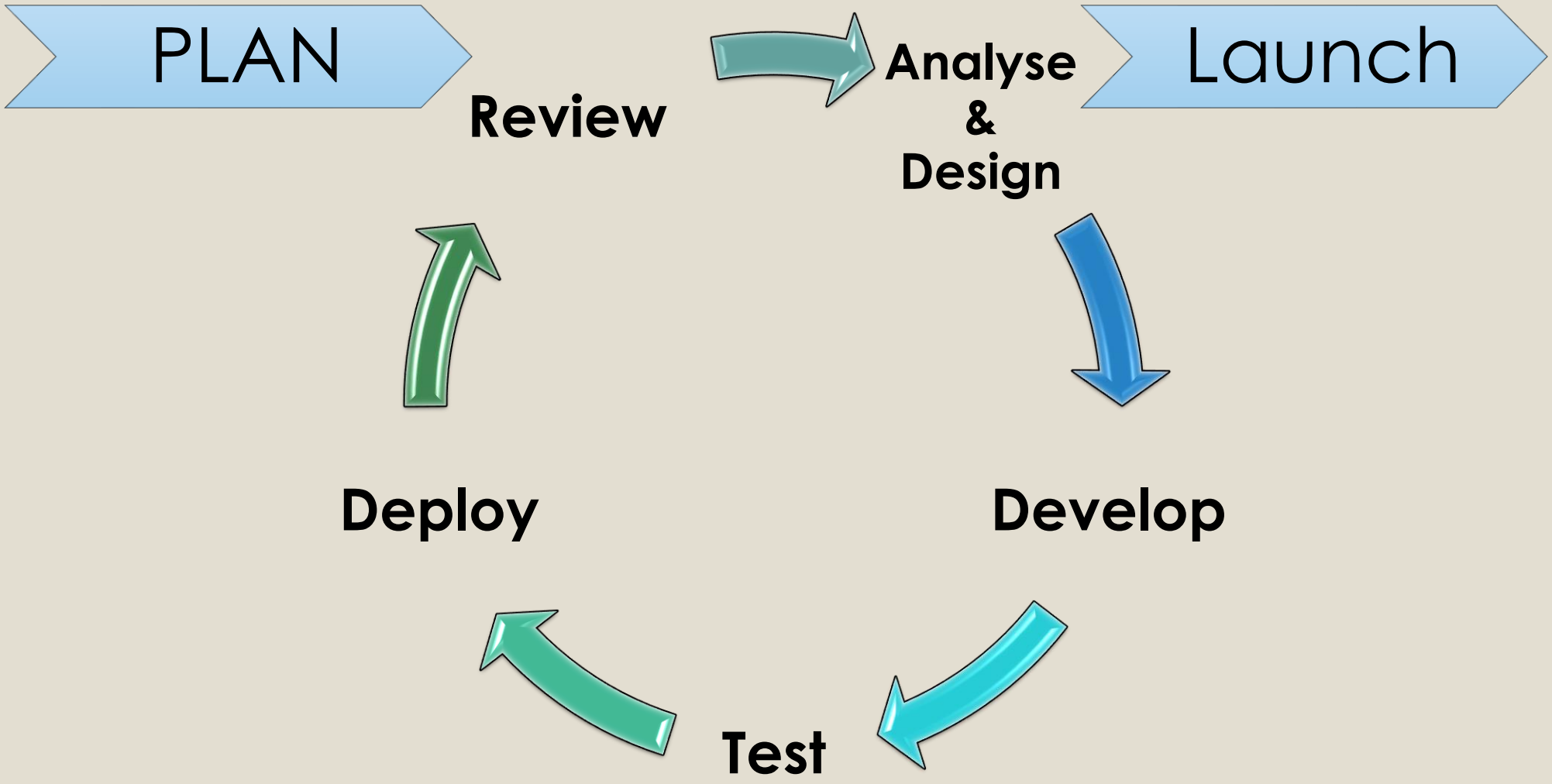


**Your turn** (3 min)

It can be:

- A memory
- A systematic example
- A fantasy situation

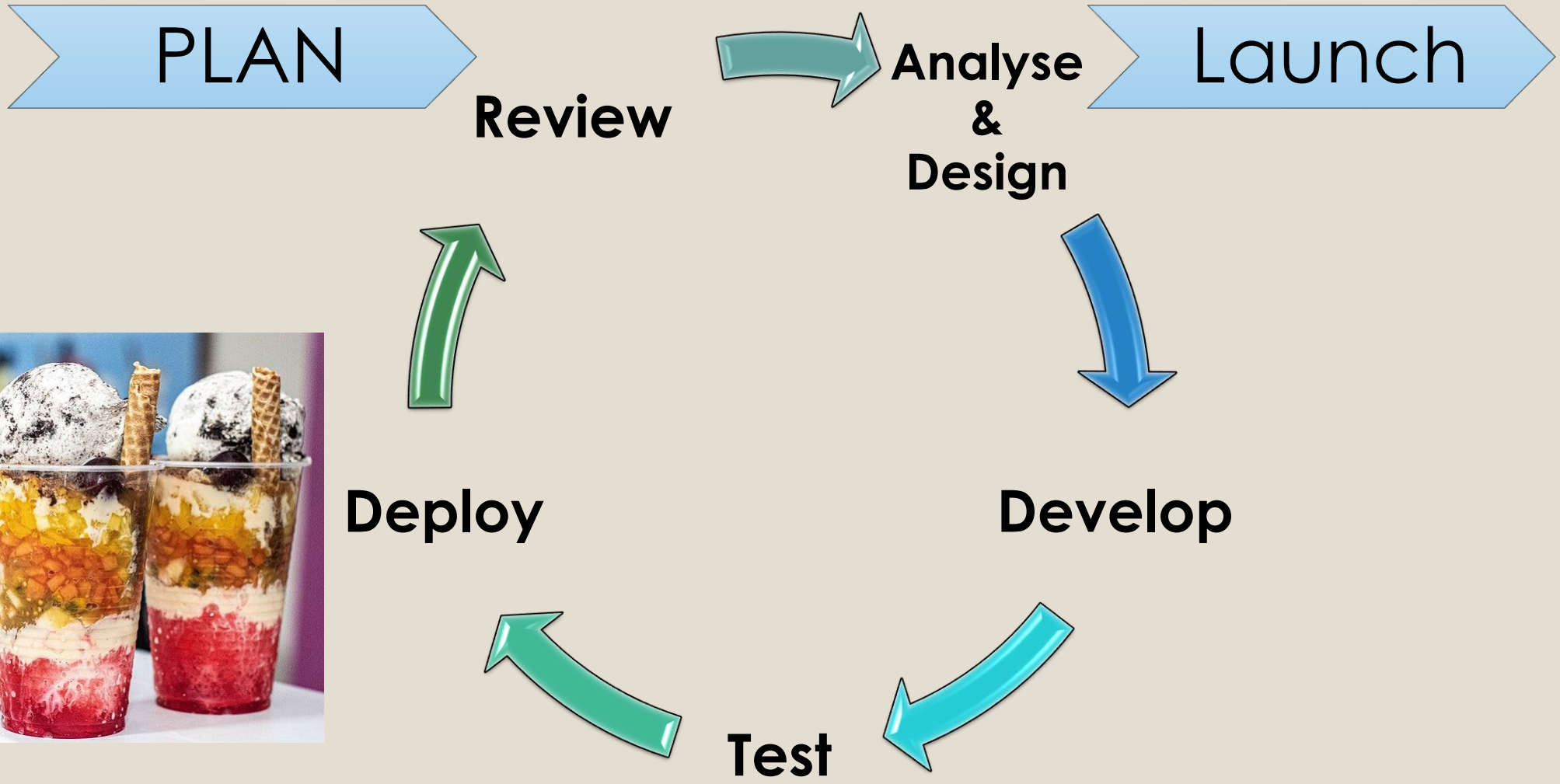
# Agile Software Development



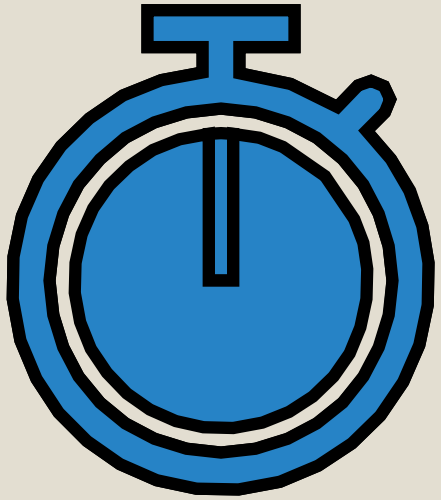
# Agile Software Development

- Dividing the work into several phases
- Incremental and iterative
- After each iteration => Inspect and adapt
- Fail early and often
- Flexible

# Real-world example of AGILE



# Real-world example of AGILE



**Your turn** (3 min)

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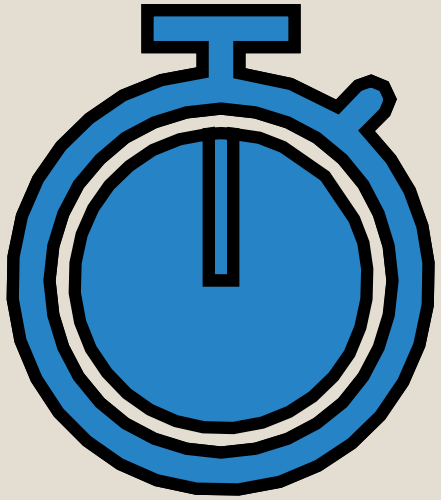
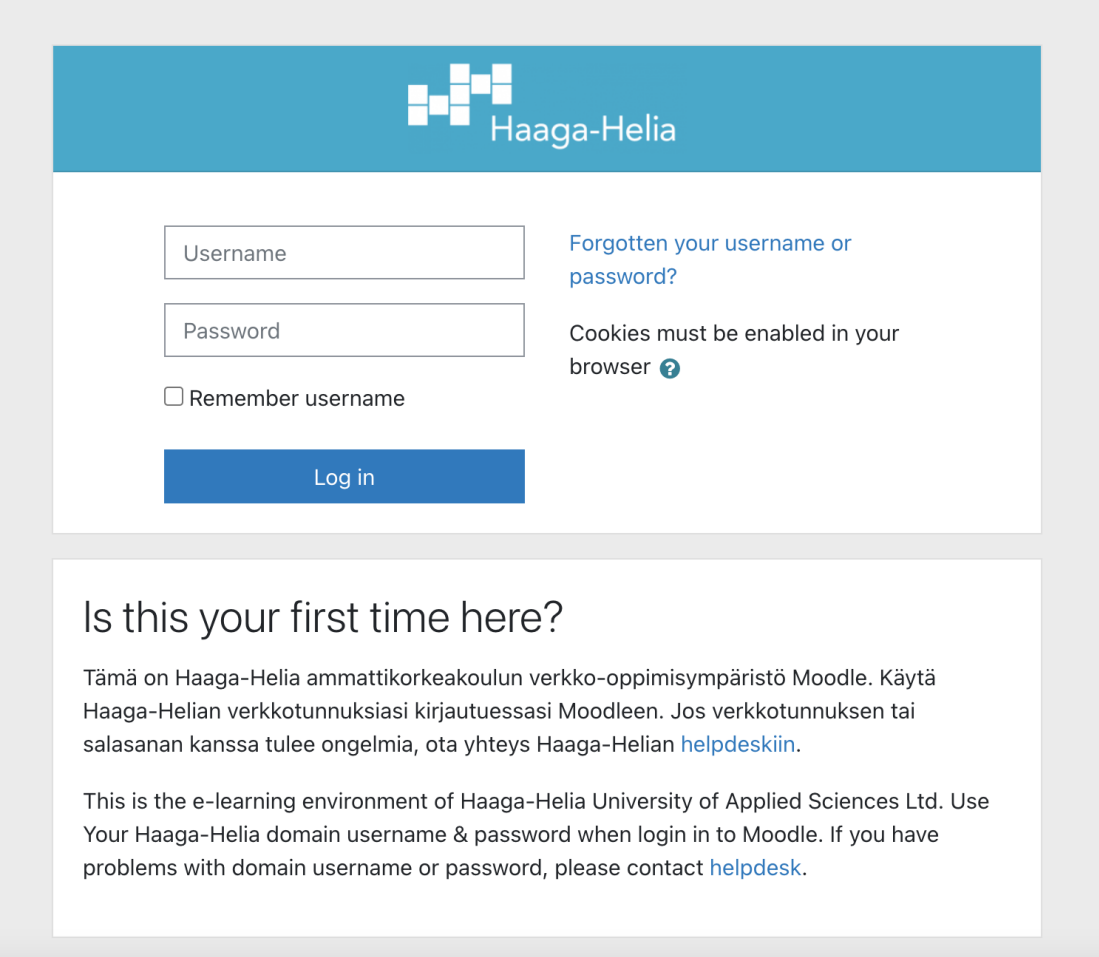
# Creating this webpage using AGILE



## Your turn

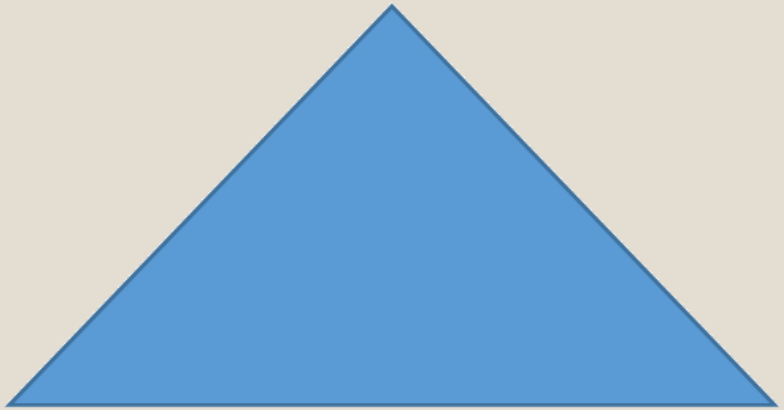
(5 min)



# How agile solves the SDLC constraints

Fixed

Time Schedule



Budget

Fixed

(Feature) Scope

Not fixed

# 12 Principals of Agile [1]

| | | |
|---|---|---|
| The customer's satisfaction is the highest priority. | Changes are welcome, at any time. | Deliver working software on a regular basis, later it can be improved. |
| Business people and developers must collaborate on a daily basis. | Hire people who are passionate about what they're doing. | Face-to-face interaction is preferred. |
| Working software is the most important indicator of progress. | The development should proceed sustainably. | The focus is on quality code and design. |
| Simplicity is essential. | Form a self-organizing team. | Reflect regularly on what is learned from successes and failures. |

# Agile event - daily standup

- Daily meeting (People who are working on a project)
- Duration: 5 – 15 min
- Everybody shortly reflect on;
  - What I did yesterday!
  - What I will do today!
  - What are obstacles!

# Agile artifact - product backlog

- List of tasks (items) that should be done, including:
  - new features
  - changes to existing features
  - bug fixes

- One item can be breakdown to multiply items!

- At each phase, every team member, selects an item from backlog and start working on it.

# Agile artifact - Sprint

- A time limit within which certain work must be done and prepared for review (1w, 2w, 1m, ..)
- Starts with plan meeting => assign tasks to people
- Ends with reviewing the performance
- Refers to the iterations!

# Agile Role – Product owner

- Translate the customer's needs to backlog items
- Prioritize the backlog items
- Approve the performance
- Responsible for customers' satisfaction

# Agile as an umbrella

**Methodology**

AGILE

**Frameworks**

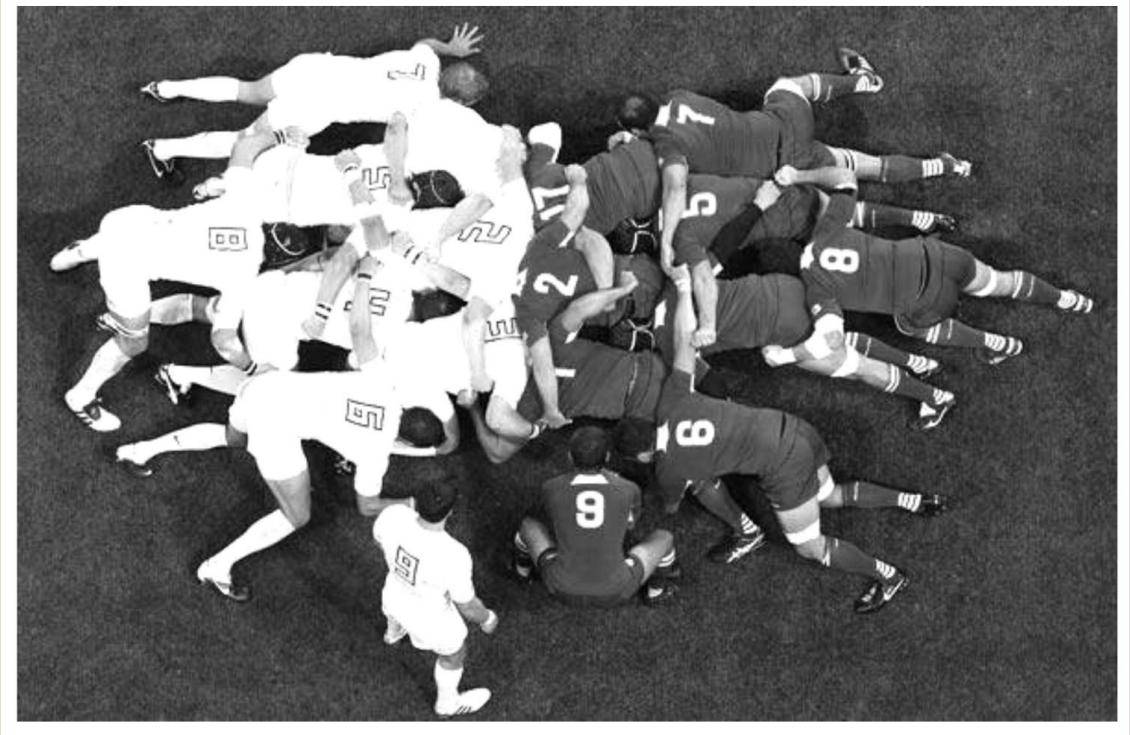
XP

SCRUM

KANBAN

# Scrum [2]

- Professional teamworking
- Strong organization ability
- Scrum team
  - **Product owner**
    - What to build
  - **Developers**
    - How to build
  - **Scrum Master**



# Scrum Role – Scrum master

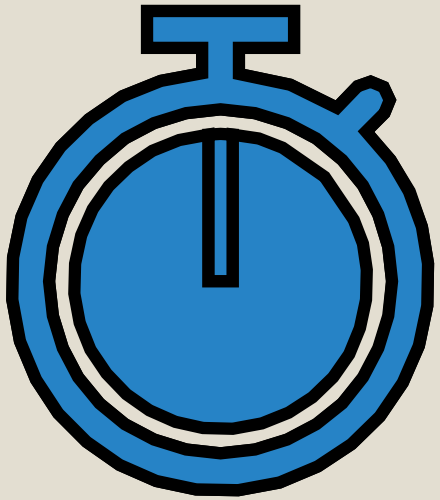
- Coordinating team activities
- Enforcing scrum roles
- Holds daily meetings
- Removes obstacles
- Not the boss but a facilitator

# Let's practice (Ball Point game)[3]

- **Material**: balls and paperboard
- **Goal**: To deliver a maximum number of balls within a **2-minute** period
- **Roles**:
  - Each team has 30-40 hand-sized balls
  - A ball is delivered if
    - It is touched by every team member
    - Every person should do some tricks or playing with the ball.
    - First and last person are fixed
- **Team leader** counts the balls and records on the paper.
- After each iteration one minute **discussion time** for reviewing and improvement insights.
- There are **Four** iterations in total!!

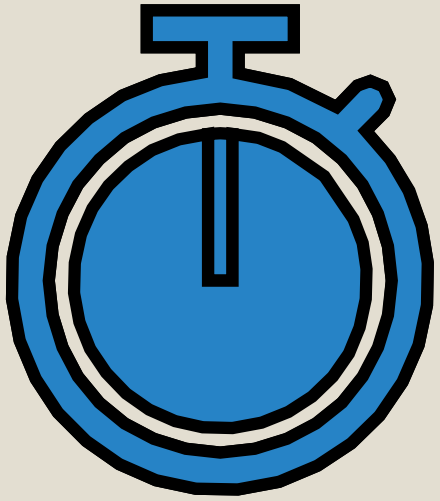
# Let's practice (Ball Point game)

- **Please get prepared ☺**
  - Position of team members,
  - Position of first and last person,
  - Position of balls and baskets,
  - Paperboard sample:



| Name of team : | | |
|---|---|---|
| Iteration | estimated | Performance |
| 1 | ?? | ?? |
| 2 | | |
| 3 | | |
| 4 | | |

# Let's practice (Ball Point game)



- **Analysis (5 min)** ☺
  - Can you find key elements of scrum:

Scrum Master Product owner

Developer team

Sprint Product backlog

Iteration Self-organizing

Reviewing

Inspect and adapt sustainable pace

Collaboration

And a lot more …[3]

# References

[1] http://agilemanifesto.org/principles.html

[2] Project management handbook, Jack C. Stanley and Erik D. Gross, https://1lib.sk/book/5599149/fc5fbc

[3] https://scrumology.com/from-the-archives-the-ball-point-game/